Software Requirements Specification

Karigar: Home Services Marketplace for Pakistan

Prepared by: Ali kazmi | Version 1.0 (Draft) | October 29, 2026 | Based on IEEE 830 and ISO/IEC/IEEE 29148

1. Introduction

1.1 Purpose

1.2 Scope

Karigar lets customers in Pakistani cities request household services (plumber, electrician, maid, AC technician, carpenter, painter) and receive price offers from verified service providers.

Many service providers (kaarigar) have limited literacy and limited smartphone use. For this reason the system follows a hybrid model: customers use a web app, dispatchers (admins) manage providers through a dashboard, and providers are reached through WhatsApp messages, voice notes and phone calls instead of through an app.

In scope for version 1.0: customer web app, admin and dispatcher dashboard, offer and quote workflow, job status tracking, ratings, complaints, notifications through WhatsApp click-to-chat links, cash payment recording.

Out of scope for version 1.0: native mobile apps, online payments, live GPS tracking, automated WhatsApp Business API messaging, provider self-service app (see Section 8).

1.3 Definitions and acronyms

Term	Definition
Customer	A person who requests a home service through the web app
Provider (kaarigar)	A plumber, electrician, maid or other skilled worker who performs the service
Dispatcher	A staff member who manages requests and contacts providers
Super admin	The business owner with full system access
Service request	A customer's request for a service, with description, address and budget
Offer	A price quote from a provider for a specific service request
RLS	Row Level Security, a database feature that limits which rows each user can read or change

Term	Definition
MVP	Minimum Viable Product, the first usable release
PKR	Pakistani Rupee

1.4 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications
- ISO/IEC/IEEE 29148:2018, Requirements engineering
- Supabase, Next.js and OpenStreetMap official documentation

1.5 Document overview

Section 2 describes the product and its users. Section 3 lists functional requirements. Sections 4 to 6 cover interfaces, quality attributes and data. Section 7 gives use cases. Sections 8 to 11 cover future scope, risks, the development plan and open issues.

2. Overall Description

2.1 Product perspective

Karigar is a new, standalone system with three parts: a customer web app, an admin and dispatcher dashboard, and a backend (database, authentication, file storage and real-time updates). WhatsApp is an external channel used to reach providers.

The core workflow has seven steps:

1. The customer submits a request with service type, address and budget.
2. The dispatcher sees the request instantly on the dashboard.
3. The dispatcher contacts suitable providers on WhatsApp or by phone.
4. Providers reply with availability and a price (text, voice note or call).
5. The dispatcher enters the offers into the system.
6. The customer compares offers and accepts one.
7. The dispatcher updates the job status until completion, then the customer rates the provider.

2.2 Product functions (summary)

- Registration, login and role-based access
- Service catalog management
- Service request creation and tracking
- Provider registration, verification and performance tracking
- Dispatch workflow and offer management
- Job status lifecycle with history
- Notifications to customers and providers
- Ratings, reviews and complaints
- Admin dashboard, reports and audit log

2.3 User classes

User class	Description	Technical skill	System access
Customer	Homeowner or tenant requesting a service	Medium	Customer web app
Provider	Skilled worker, often with low literacy	Low	None directly. Reached by WhatsApp and phone
Dispatcher	Staff member who handles requests and providers	Medium	Admin dashboard
Super admin	Business owner	Medium	Full access, staff management, reports

2.4 Operating environment

- Latest two versions of Chrome, Edge, Safari and Firefox on desktop and mobile
- Mobile-first design, minimum screen width 360 px
- Hosting on Vercel, PostgreSQL database and authentication on Supabase

2.5 Design and implementation constraints

- Providers must never be required to install an app or fill in forms
- Interface language must be simple (English with Roman Urdu labels)
- All prices are in PKR and exclusive of tax
- Payment in version 1.0 is cash on completion only
- The system must run within free or low-cost tiers of the chosen services
- Secret keys must never be placed in client-side code

2.6 Assumptions and dependencies

- Dispatchers have a smartphone or computer with internet and WhatsApp
- Providers can be reached on WhatsApp or by phone
- Supabase, Vercel and OpenStreetMap services remain available
- The initial launch targets one or two cities (for example Rawalpindi and Islamabad)

3. Functional Requirements

Priority key: **M** = Must (MVP), **S** = Should, **C** = Could.

3.1 Authentication and roles

ID	Requirement	Priority
FR-AUTH-01	The system shall allow customers to register with name, phone number and password	M
FR-AUTH-02	The system shall allow login, logout and password reset	M

ID	Requirement	Priority
FR-AUTH-03	The system shall enforce role-based access for customer, dispatcher and super admin	M
FR-AUTH-04	Staff accounts shall be created only by the super admin. Public signup shall not create staff roles	M
FR-AUTH-05	The system shall verify customer phone numbers by one-time password	S
FR-AUTH-06	Customers shall be able to edit their profile and save multiple addresses	S

3.2 Service catalog

ID	Requirement	Priority
FR-CAT-01	The system shall display service categories with an icon and short description	M
FR-CAT-02	Admins shall be able to add, edit and disable categories and sub-services	M
FR-CAT-03	The system shall show an indicative price range for each category	S

3.3 Service requests

ID	Requirement	Priority
FR-REQ-01	A customer shall be able to create a request with category, problem description, address, preferred date and time, and budget	M
FR-REQ-02	The system shall generate a unique reference number for every request (for example KH-000123)	M
FR-REQ-03	The system shall validate required fields and phone number format before saving	M
FR-REQ-04	Customers shall see a list of their requests with current status	M
FR-REQ-05	Customers shall be able to cancel a request before the job starts	M
FR-REQ-06	Customers shall be able to attach up to five photos of the problem	S
FR-REQ-07	Customers shall be able to mark their location on a map	S
FR-REQ-08	Customers shall be able to record a voice note describing the problem	C

3.4 Provider management

ID	Requirement	Priority
FR-PRV-01	Dispatchers shall register providers with name, WhatsApp number, categories, service areas and years of experience	M
FR-PRV-02	Each provider shall have a status: pending, verified or suspended	M
FR-PRV-03	Dispatchers shall be able to set a provider as available or unavailable	M
FR-PRV-04	The system shall show provider performance: jobs completed, average rating and complaints	M
FR-PRV-05	Dispatchers shall be able to search and filter providers by category, area and status	M
FR-PRV-06	The system shall store identity documents (for example CNIC copy) in private storage visible only to staff	S

3.5 Dispatch and offers

ID	Requirement	Priority
FR-DSP-01	New requests shall appear on the dispatcher dashboard in real time without a page refresh	M
FR-DSP-02	The system shall suggest verified, available providers that match the request category and area	M
FR-DSP-03	The system shall generate a pre-filled WhatsApp message link containing the request details for a chosen provider	M
FR-DSP-04	Dispatchers shall be able to record a provider's reply (accepted or declined) and quoted price	M
FR-DSP-05	A request shall be able to receive multiple offers	M
FR-DSP-06	Customers shall see each offer with provider name, rating, price and expected arrival time	M
FR-DSP-07	When a customer accepts one offer, the system shall close all other offers automatically	M
FR-DSP-08	Offers shall expire after a configurable time (default 30 minutes)	S

3.6 Job lifecycle

ID	Requirement	Priority

ID	Requirement	Priority
FR-JOB-01	The system shall support these statuses: New, Dispatching, Offers received, Confirmed, On the way, In progress, Completed, Cancelled	M
FR-JOB-02	Every status change shall be saved with time and the user who made it	M
FR-JOB-03	Customers shall see the live status of their job	M
FR-JOB-04	On completion, the final price and cash payment shall be recorded	M
FR-JOB-05	Dispatchers shall be able to reassign a job to another provider if the first provider does not arrive	M

3.7 Notifications

ID	Requirement	Priority
FR-NTF-01	The system shall notify customers inside the app when offers arrive and when status changes	M
FR-NTF-02	The system shall provide the dispatcher with a WhatsApp message for the provider on confirmation	M
FR-NTF-03	The system shall send customers a confirmation by SMS or WhatsApp	S
FR-NTF-04	The system shall keep a log of notifications sent	S

3.8 Ratings and reviews

ID	Requirement	Priority
FR-RAT-01	Customers shall be able to rate a completed job from 1 to 5 and add a comment	M
FR-RAT-02	Only one review shall be allowed per completed job	M
FR-RAT-03	The system shall recalculate the provider's average rating after each review	M
FR-RAT-04	Admins shall be able to hide abusive reviews	S

3.9 Complaints

ID	Requirement	Priority
FR-CMP-01	Customers shall be able to file a complaint about a job	S

ID	Requirement	Priority
FR-CMP-02	Dispatchers shall be able to manage complaints with status open, in review and resolved	S
FR-CMP-03	The system shall flag providers who receive repeated complaints	S

3.10 Admin dashboard and reports

ID	Requirement	Priority
FR-ADM-01	The dashboard shall show requests today, active jobs, completed jobs and cancellations	M
FR-ADM-02	Staff shall be able to search and filter requests by status, date, category and area	M
FR-ADM-03	The system shall export request data as CSV	S
FR-ADM-04	The system shall show charts of weekly requests and completions by category	S
FR-ADM-05	The system shall keep an audit log of important staff actions	S

4. External Interface Requirements

4.1 User interfaces

- Responsive layout that works on phones, tablets and desktops
- Icons together with short text labels for every service and action
- Large tap targets (at least 44 px) and clear contrast (WCAG AA)
- Language: English with Roman Urdu labels. Urdu script is planned for a later version

4.2 Hardware interfaces

No special hardware. Users need a device with a modern browser and an internet connection.

4.3 Software interfaces

Interface	Purpose
Supabase (PostgreSQL, Auth, Storage, Realtime)	Data, login, photo storage and live updates
WhatsApp click-to-chat links (wa.me)	Reaching providers in version 1.0
WhatsApp Business Cloud API	Automated messages in a future version
OpenStreetMap and Leaflet	Map display and location selection
Vercel	Hosting and deployment

4.4 Communications interfaces

All traffic shall use HTTPS (TLS 1.2 or higher). Live dashboard updates shall use WebSocket connections.

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Pages shall load in under 3 seconds on a 4G connection. Dashboard updates shall appear within 2 seconds
NFR-02	Capacity	The system shall support 500 concurrent users and 10,000 requests per month
NFR-03	Availability	Target uptime of 99% per month
NFR-04	Security	Passwords shall be hashed by the authentication provider. Row Level Security shall ensure customers can only read their own data. Staff permissions shall be enforced on the server. Inputs shall be validated against injection and script attacks
NFR-05	Privacy	A provider's phone number shall not be shown to customers until an offer is accepted. A customer's contact details shall be shared with a provider only after confirmation. Only necessary personal data shall be collected
NFR-06	Usability	A new customer shall be able to submit a request in under 2 minutes without help
NFR-07	Localization	Prices in PKR, dates and times in Asia/Karachi time zone
NFR-08	Maintainability	Source code shall be in version control, modular and documented
NFR-09	Backup	Daily automatic database backups
NFR-10	Compatibility	Support the latest two versions of major browsers
NFR-11	Auditability	Status changes and staff actions shall be traceable to a user and time

6. Data Requirements

6.1 Main entities

Entity	Key attributes
profiles	id, full name, phone, role, created at
categories	id, name, icon, description, active, price range

Entity	Key attributes
providers	id, name, WhatsApp number, status, available, areas, experience, average rating
provider_categories	provider id, category id
service_requests	id, reference number, customer id, category id, description, address, latitude, longitude, preferred time, budget, status
offers	id, request id, provider id, price, arrival time, status, expires at
job_events	id, request id, status, changed by, time, note
reviews	id, request id, provider id, customer id, rating, comment, hidden
complaints	id, request id, customer id, text, status
notifications	id, user id, message, channel, sent at
audit_logs	id, staff id, action, target, time

6.2 Request status lifecycle

New, then Dispatching, then Offers received, then Confirmed, then On the way, then In progress, then Completed. A request can move to Cancelled from any state before In progress.

7. Use Cases

ID	Name	Actor	Summary
UC-01	Submit service request	Customer	Customer logs in, selects a service, enters problem, address and budget, and submits. The system creates a request with a reference number
UC-02	Dispatch to providers	Dispatcher	Dispatcher opens a new request, reviews suggested providers, sends WhatsApp messages, and records replies and prices as offers
UC-03	Accept offer	Customer	Customer compares offers by price and rating and accepts one. Other offers close and the job becomes Confirmed
UC-04	Update job status	Dispatcher	After calling the provider, dispatcher updates status to On the way, In progress or Completed. The customer sees the change live
UC-05	Rate provider	Customer	After completion, customer gives a 1 to 5 rating and an optional comment. Average rating updates

ID	Name	Actor	Summary
UC-06	Register provider	Dispatcher	Dispatcher enters provider details by phone call, sets categories and areas, and marks the provider verified after checks

8. Out of Scope and Future Enhancements

- Native Android and iOS apps (installable web app as an intermediate step)
- Online payments (JazzCash, Easypaisa, cards) and commission handling
- Live GPS tracking of providers
- Automated WhatsApp Business API messaging and chatbot
- Provider self-service portal with voice-based interface
- Urdu (Nastaliq) interface and Urdu voice notes transcription
- Subscription plans, promotions and referral rewards

9. Risks and Mitigation

ID	Risk	Mitigation
R1	Providers do not respond or are hard to reach	Contact several providers per request. Track response rate and prefer reliable providers
R2	Provider does not show up	Reassignment feature, ratings and complaint flagging
R3	Customers and providers deal directly and bypass the platform	Share contact details only after confirmation. Add value through ratings and guarantees
R4	Trust and safety concerns about strangers entering homes	Provider verification with identity documents, reviews and a complaint process
R5	Dependence on WhatsApp policies	Keep click-to-chat as a fallback and plan the Business API for later
R6	Running costs grow with usage	Use free tiers first, monitor usage and set alerts
R7	Misuse of personal data	Row Level Security, private storage, minimal data collection

10. Development Plan and Acceptance Criteria

10.1 Phases

Phase	Content	Estimated time
1	Project setup, authentication and roles	1 week

Phase	Content	Estimated time
2	Service catalog and request creation	1 to 2 weeks
3	Provider management and dispatch dashboard with real-time updates	2 weeks
4	Offers, acceptance and job lifecycle	1 to 2 weeks
5	Ratings, complaints, reports and polish	1 to 2 weeks
6	Testing, deployment and demo data	1 week

10.2 MVP acceptance criteria

- A customer can register, submit a request and track it to completion
- A dispatcher sees a new request within 2 seconds without refreshing
- A request can receive at least three offers and the customer can accept one
- Every status change is stored with time and user
- A customer can only view their own requests (verified by testing with two accounts)
- All Must requirements in Section 3 pass testing on a mobile phone and a desktop browser
- The system is deployed publicly with demo accounts and sample data

11. Open Issues

- Final product name (Karigar is a working title)
- Launch cities and the service categories to start with
- Business model: commission per job, subscription or listing fee
- Whether to adopt the WhatsApp Business API in version 1.1
- Identity verification process for providers

Revision History

Version	Date	Author	Description
1.0	October 29, 2026	Ali kazmi	Initial draft